–Supplemental Material– Prune but Preserve: Decoupling Redundancy Suppression and Semantic Preservation for Efficient Time Series Clustering

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I. Baseline Descriptions

The baseline set is organized around the empirical question studied in the main paper, namely at which stage redundancy should be reduced for unsupervised MTS clustering. It covers five streams: full-sequence representation learning, input-side token exclusion, partial-observation clustering, structural subsequence indexing, and foundation-style time series modeling. This design separates the benefit of source-level patch admission from gains brought by contrastive objectives, spectral modeling, graph structure, incomplete-view learning, or general temporal pretraining. For embedding-output baselines, k-means gives final assignments with the label-provided cluster number k, following the same unsupervised protocol as the main paper.

FCACC [1] is a fuzzy cluster-aware contrastive clustering method for time series. It constructs augmented views and introduces cluster-aware hard negatives to reduce the gap between instance contrast and cluster assignment. FCACC represents a clustering-oriented full-sequence baseline, testing whether cluster-aware contrast can reduce background influence without source-level patch admission. The official implementation is available at FCACC.

TFMCC [2] performs time-frequency augmented multi-level contrastive clustering. Temporal and spectral augmentations enhance cluster-relevant patterns at different representation levels, but the complete sequence still enters the encoding pipeline. TFMCC tests whether temporal-spectral contrast can handle redundancy as effectively as pruning before encoder-side interaction. The official implementation is available at TFMCC.

GTM [3] learns general time series representations through frequency-domain attention and shared temporal modeling. As GTM is designed for transferable representation rather than clustering-specific reduction, k-means is applied to its embeddings for final assignment. The comparison examines whether broad temporal pretraining can replace explicit patch admission under redundancy-dominated MTS records. The official implementation is available at GTM.

EMTC [4] is the closest input-side exclusion baseline for MTS clustering. It suppresses redundancy through importance-aware variate-wise masking and enriches retained content with multi-endogenous views. Its comparison with SIGMA directly tests whether confidence-calibrated admission, routed semantic restoration, and prototype-guided refinement improve clustering after sparse token admission. The official implementation is available at EMTC.

FEI [5] introduces a frequency-domain information bottleneck by masking frequency components and predicting the corresponding embeddings. This design encourages informative spectral structure without manually defined positive and negative pairs. FEI serves as a spectral-reduction baseline, assessing whether patch-level admission still helps beyond frequency-oriented representation learning. The official implementation is available at FEI.

MVCIMTS [6] is a contrastive multi-view clustering method for incomplete MTS. It treats variates as views and learns complementary information through multi-view contrast. Since its partial observation comes from variables instead of temporal patches, MVCIMTS tests whether variable-wise incompleteness brings the same clustering benefit as temporal admission followed by routed restoration. The official implementation is available at MVCIMTS.

k-Graph [7] converts subsequences into graph representations and clusters graph-derived features. By building graphs across multiple subsequence lengths, k-Graph captures recurring local patterns and structural similarity. It provides a data-centric baseline for datasets whose clusters depend on shape correspondence or repeated subsequence structure. The official implementation is available at k-Graph.

TimesURL [8] is a universal self-supervised representation method for time series. It combines contrastive learning with temporal constraints to obtain general embeddings. In the prune-and-preserve comparison, TimesURL represents a full-sequence self-supervised baseline because redundancy is not controlled through an admissible patch set. The official implementation is available at TimesURL.

UNITS [9] is a unified multi-task time series model. It represents task information as tokens in a shared modeling framework and learns transferable representations across domains, temporal scales, and sampling patterns. UNITS provides a foundation-style baseline for testing whether task-unified pretraining is sufficient for redundancy-dominated MTS clustering. Its comparison with SIGMA highlights the role of source-level admission and routed semantic preservation in cluster-oriented representation learning. The official implementation is available at UNITS.

II. Implementation Details

All experiments follow the unsupervised clustering protocol in the main paper. For each UEA dataset, training and test splits are concatenated before clustering, and labels are accessed only for metric computation. Each experiment is repeated with five random seeds, and the mean and standard deviation are reported. This protocol keeps the supplemental material consistent with the main-paper evaluation and makes the supplementary tables comparable to the representative results in Section IV of the full paper.

SIGMA uses Confidence-Calibrated Masked Admission as the default admission strategy. In the masking ablation, source-level admission is removed while all other settings remain unchanged, isolating the effect of controlling patch participation before encoder-side feature interaction. Unless otherwise specified, no input perturbation is added, and Gaussian noise is introduced only in the robustness study.

For patch-level admission, the patch length is set to $\ell_{\mathrm{p}} = 4$. Given $m = \lceil T/\ell_{\mathrm{p}} \rceil$ patches, the retained ratio is $r_{\mathrm{a}} = 0.5$, and the retained patch number is $m_{\mathrm{a}} = \lceil r_{\mathrm{a}} m \rceil$. Thus, the encoder processes a compact patch subset after admission. This setting reduces redundant patch participation before feature interaction while keeping enough temporal context for semantic restoration.

For endogenous-view representation learning, the number of views is $n_{\mathrm{v}} = 2$. Each view encoder uses hidden dimension $d_{\mathrm{h}} = 32$, and the fused clustering embedding has dimension $d_{\mathrm{z}} = 64$. The prototype alignment temperature is $\tau_{\mathrm{p}} = 0.1$. These values give SIGMA a compact representation budget while allowing admitted content to be organized into complementary latent views.

SIGMA is trained for $e_{\max} = 200$ epochs with mini-batch size $b = 64$. Adam is adopted with learning rate $\alpha = 0.001$. The loss weights are $\lambda_{\mathrm{rec}} = 1.0$, $\lambda_{\mathrm{orth}} = 0.1$, and $\lambda_{\mathrm{pa}} = 1.0$, corresponding to reconstruction, inter-view orthogonality, and prototype alignment, respectively. Reconstruction anchors temporal content after admission, orthogonality discourages view homogenization, and prototype alignment sharpens the fused cluster space. Under this configuration, admission control, semantic restoration, and cluster refinement keep distinct roles within the prune-and-preserve pipeline.

III. Supplementary Results

This section provides the supplementary empirical record behind the main paper. The main paper keeps the evaluation compact through representative datasets and condensed diagnostics. The supplemental material reports complete benchmark tables, Bonferroni-Dunn rank tests, full module and route ablations, additional training-loss curves, mask-rate scalability, and the robustness and compactness diagnostics in Fig. 4 under the same unsupervised clustering protocol defined in Section II. These results do not introduce another evaluation setting. They make the main claims traceable at the dataset, component, optimization, statistical, and resource levels.

The organization follows the role assignment in SIGMA. Complete clustering results and rank tests examine whether source-level admission and routed representation learning remain competitive across heterogeneous MTS collections. Ablations verify the contributions of admission, endogenous-view organization, semantic restoration, and prototype-guided refinement. Loss curves inspect whether staged training stays controlled after prototype guidance is introduced. Mask-rate scalability checks whether runtime reduction comes from fewer admitted patches before encoder-side interaction. Robustness and compactness diagnostics further test whether the retained representation remains usable under noise perturbation and reduced embedding dimensions. Together, these supplementary results connect clustering quality, statistical reliability, mechanism attribution, training stability, and computation-aware temporal data organization.

A. Complete Clustering Results on 30 UEA Datasets

The main paper answers Q1 with representative per-dataset results and a Bonferroni-Dunn rank test. This supplemental material subsection provides the complete dataset-level record. TABLE I and TABLE II report ACC, F1, NMI, and ARI on all 30 UEA datasets, covering both the datasets discussed in the main paper and the remaining benchmarks omitted due to page limits. The supplemental material therefore supports Q1 with the full benchmark evidence under the same unsupervised protocol.

The complete tables show that SIGMA's advantage does not come from a few favorable cases. On datasets with clear cluster-revealing patterns, such as ArticularyWordRecognition, BasicMotions, JapaneseVowels, PenDigits, and SelfRegulationSCP1, SIGMA obtains leading or highly competitive scores across multiple metrics. These datasets are important for the prune-and-preserve claim, because sparse class-related changes can be diluted when complete sequences enter feature interaction. By reducing background-dominated patch participation before encoding and organizing retained content through routed objectives, SIGMA keeps compact representations informative for clustering. JapaneseVowels and PenDigits are typical cases where reduced input participation still yields a discriminative grouping space.

The full benchmark also reflects the diversity of unsupervised MTS clustering. Several datasets favor specialized baselines when cluster similarity depends more on shape matching,

TABLE I

COMPLETE CLUSTERING RESULTS ON 30 UEA DATASETS EVALUATED BY ACC, F1, NMI, AND ARI, PART I. THE **BEST** AND **SECOND BEST** RESULTS ARE HIGHLIGHTED BY CELL BACKGROUND. OOM DENOTES OUT-OF-MEMORY RUNS.

Datasets	Metric	FCACC (PR'26)	GTM (ICLR'26)	EMTC (AAAI'26)	TFMCC (AAAI'26)	FEI (AAAI'25)	MVCIMTS (INFFUS'25)	k-Graph (TKDE'25)	TimesURL (AAAI'24)	UNITS (NeurIPS'24)	SIGMA (ours)
ArticularyWordRecognition	ACC	0.7329 ± 0.0037	0.3555 ± 0.0380	0.3547 ± 0.0185	0.5030 ± 0.0253	0.4710 ± 0.0731	0.0497 ± 0.0124	0.2838 ± 0.0159	0.6929 ± 0.0516	0.2948 ± 0.0200	0.7603 ± 0.0232
	F1	0.8039 ± 0.0055	0.3526 ± 0.0348	0.3560 ± 0.0214	0.3994 ± 0.0241	0.4974 ± 0.0815	0.0109 ± 0.0096	0.2774 ± 0.0169	0.7425 ± 0.0413	0.3114 ± 0.0243	0.7929 ± 0.0392
	NMI	0.8408 ± 0.0064	0.4708 ± 0.0311	0.5070 ± 0.0272	0.7012 ± 0.0159	0.6185 ± 0.0744	0.0314 ± 0.0428	0.3948 ± 0.0099	0.8215 ± 0.0145	0.4280 ± 0.0198	0.8276 ± 0.0165
	ARI	0.6641 ± 0.0045	0.1833 ± 0.0296	0.1615 ± 0.0271	0.4466 ± 0.0210	0.3352 ± 0.0843	0.0019 ± 0.0031	0.1224 ± 0.0091	0.6214 ± 0.0340	0.1395 ± 0.0188	0.6466 ± 0.0257
AtrialFibrillation	ACC	0.4333 ± 0.0000	0.4667 ± 0.0422	0.6133 ± 0.0499	0.4333 ± 0.0558	0.3867 ± 0.0163	0.4095 ± 0.0440	0.4400 ± 0.0389	0.4600 ± 0.0490	0.4500 ± 0.0167	0.5000 ± 0.0365
	F1	0.4303 ± 0.0028	0.4593 ± 0.0393	0.5828 ± 0.0633	0.4041 ± 0.0191	0.3543 ± 0.0194	0.3491 ± 0.0555	0.4295 ± 0.0406	0.4169 ± 0.0278	0.4343 ± 0.0089	0.4956 ± 0.0778
	NMI	0.1005 ± 0.0040	0.0726 ± 0.0277	0.2883 ± 0.0758	0.0544 ± 0.0351	0.0353 ± 0.0237	0.1445 ± 0.0821	0.1143 ± 0.0748	0.0858 ± 0.0339	0.0643 ± 0.0072	0.1251 ± 0.0390
	ARI	-0.0056 ± 0.0011	0.0051 ± 0.0258	0.1248 ± 0.0788	0.0064 ± 0.0282	-0.0360 ± 0.0089	0.1058 ± 0.0600	0.0039 ± 0.0462	0.0136 ± 0.0342	-0.0125 ± 0.0012	0.0582 ± 0.0393
BasicMotions	ACC	0.6675 ± 0.0100	0.7300 ± 0.1086	0.7950 ± 0.0292	0.9850 ± 0.0122	0.6775 ± 0.0443	0.2975 ± 0.0827	0.8875 ± 0.1321	0.7700 ± 0.1582	0.5437 ± 0.0812	0.9875 ± 0.0079
	F1	0.8228 ± 0.0098	0.7311 ± 0.1060	0.7832 ± 0.0275	0.9701 ± 0.0235	0.7351 ± 0.0723	0.1517 ± 0.0794	0.8792 ± 0.1500	0.8627 ± 0.0835	0.5705 ± 0.0412	0.9875 ± 0.0079
	NMI	0.8048 ± 0.0020	0.5569 ± 0.1633	0.6949 ± 0.0711	0.9627 ± 0.0248	0.6792 ± 0.0317	0.1007 ± 0.1775	0.8626 ± 0.0575	0.8487 ± 0.0634	0.3898 ± 0.1107	0.9665 ± 0.0191
	ARI	0.6295 ± 0.0045	0.4526 ± 0.1902	0.6085 ± 0.0606	0.9606 ± 0.0311	0.5317 ± 0.0351	0.0690 ± 0.1381	0.8174 ± 0.1218	0.7349 ± 0.1448	0.2989 ± 0.1276	0.9666 ± 0.0208
CharacterTrajectories	ACC	0.7981 ± 0.0012	0.0651 ± 0.0000	0.3000 ± 0.0261	0.3677 ± 0.0448	0.6040 ± 0.0096	0.0866 ± 0.0318	0.4500 ± 0.0167	0.0651 ± 0.0000	0.4533 ± 0.0267	0.7668 ± 0.0529
	F1	0.8583 ± 0.0012	0.0061 ± 0.0000	0.2886 ± 0.0434	0.3405 ± 0.0281	0.6289 ± 0.0099	0.0196 ± 0.0166	0.4350 ± 0.0149	0.1222 ± 0.0000	0.1703 ± 0.0373	0.8019 ± 0.0378
	NMI	0.8726 ± 0.0013	0.0000 ± 0.0000	0.3633 ± 0.0432	0.5756 ± 0.0418	0.7095 ± 0.0041	0.0682 ± 0.1107	0.5347 ± 0.0121	0.0000 ± 0.0000	0.1776 ± 0.0688	0.8261 ± 0.0315
	ARI	0.7705 ± 0.0012	0.0000 ± 0.0000	0.1756 ± 0.0296	0.1913 ± 0.0474	0.5137 ± 0.0138	0.0181 ± 0.0345	0.3211 ± 0.0193	0.0000 ± 0.0000	0.0532 ± 0.0279	0.6933 ± 0.0563
Cricket	ACC	0.9407 ± 0.0288	0.4533 ± 0.0264	0.5528 ± 0.0204	0.6922 ± 0.0664	0.5744 ± 0.0834	0.1167 ± 0.0350	0.4589 ± 0.0218	0.6722 ± 0.0813	0.4922 ± 0.0990	0.8489 ± 0.0787
	F1	0.9599 ± 0.0024	0.4393 ± 0.0376	0.5635 ± 0.0383	0.6319 ± 0.0703	0.6027 ± 0.0590	0.0417 ± 0.0250	0.4545 ± 0.0225	0.7562 ± 0.0557	0.5414 ± 0.0992	0.8838 ± 0.0523
	NMI	0.9509 ± 0.0020	0.5189 ± 0.0101	0.6181 ± 0.0286	0.7858 ± 0.0464	0.6794 ± 0.0579	0.0817 ± 0.0964	0.4752 ± 0.0228	0.7958 ± 0.0391	0.5947 ± 0.0841	0.8764 ± 0.0455
	ARI	0.9051 ± 0.0158	0.2584 ± 0.0194	0.2971 ± 0.0365	0.5842 ± 0.0827	0.4468 ± 0.0872	0.0171 ± 0.0299	0.2533 ± 0.0250	0.6028 ± 0.0783	0.3359 ± 0.1072	0.7672 ± 0.0967
DuckDuckGeese	ACC	0.3067 ± 0.0125	0.4140 ± 0.0162	0.4240 ± 0.0408	0.4720 ± 0.0172	0.3220 ± 0.0117	0.2620 ± 0.0519	0.3280 ± 0.0160	0.3220 ± 0.0147	0.2960 ± 0.0136	0.4020 ± 0.0325
	F1	0.2611 ± 0.0200	0.4042 ± 0.0246	0.4003 ± 0.0494	0.3394 ± 0.0193	0.3013 ± 0.0347	0.1650 ± 0.0575	0.3017 ± 0.0107	0.2641 ± 0.0182	0.2869 ± 0.0075	0.3819 ± 0.0291
	NMI	0.0800 ± 0.0179	0.2483 ± 0.0205	0.2082 ± 0.0380	0.2791 ± 0.0267	0.1030 ± 0.0090	0.0335 ± 0.0301	0.0958 ± 0.0101	0.1732 ± 0.0351	0.0682 ± 0.0175	0.1718 ± 0.0577
	ARI	0.0227 ± 0.0109	0.1394 ± 0.0210	0.0729 ± 0.0331	0.1780 ± 0.0248	0.0369 ± 0.0091	0.0073 ± 0.0170	0.0225 ± 0.0087	0.0375 ± 0.0110	0.0064 ± 0.0105	0.1000 ± 0.0580
EigenWorms	ACC	OOM	0.3876 ± 0.0072	OOM	OOM	0.4185 ± 0.0361	0.4324 ± 0.0024	0.3135 ± 0.0236	0.3660 ± 0.0510	0.4525 ± 0.0366	0.4332 ± 0.0402
	F1	OOM	0.2868 ± 0.0068	OOM	OOM	0.3899 ± 0.0605	0.1545 ± 0.0173	0.2920 ± 0.0180	0.3035 ± 0.0806	0.4410 ± 0.0601	0.4016 ± 0.0483
	NMI	OOM	0.0683 ± 0.0035	OOM	OOM	0.1826 ± 0.0152	0.0300 ± 0.0094	0.0848 ± 0.0181	0.0697 ± 0.0572	0.2872 ± 0.0919	0.1885 ± 0.0300
	ARI	OOM	0.0626 ± 0.0052	OOM	OOM	0.1453 ± 0.0227	0.0098 ± 0.0046	0.0400 ± 0.0160	0.0515 ± 0.0699	0.1694 ± 0.0486	0.1620 ± 0.0458
Epilepsy	ACC	0.4279 ± 0.0297	0.3869 ± 0.0404	0.5203 ± 0.0244	0.4953 ± 0.0477	0.6364 ± 0.0808	0.2953 ± 0.0324	0.6167 ± 0.0316	0.5156 ± 0.0189	0.4982 ± 0.0646	0.4756 ± 0.0321
	F1	0.4295 ± 0.0302	0.3710 ± 0.0430	0.5136 ± 0.0265	0.4147 ± 0.0552	0.6200 ± 0.0821	0.1882 ± 0.0235	0.6200 ± 0.0260	0.4979 ± 0.0080	0.5086 ± 0.0930	0.4724 ± 0.0357
	NMI	0.1397 ± 0.0347	0.1015 ± 0.0345	0.2476 ± 0.0419	0.3029 ± 0.0794	0.4079 ± 0.0773	0.0205 ± 0.0340	0.4013 ± 0.0415	0.3363 ± 0.0795	0.3126 ± 0.0851	0.2652 ± 0.0289
	ARI	0.0960 ± 0.0211	0.0814 ± 0.0434	0.1803 ± 0.0389	0.1992 ± 0.0638	0.3829 ± 0.0804	0.0076 ± 0.0184	0.3275 ± 0.0526	0.2288 ± 0.0542	0.2320 ± 0.0733	0.1889 ± 0.0197
ERing	ACC	0.6327 ± 0.0360	0.4180 ± 0.0585	0.6704 ± 0.0486	0.8607 ± 0.0218	0.7120 ± 0.0918	0.3667 ± 0.1560	0.5380 ± 0.0421	0.7433 ± 0.1161	0.4087 ± 0.0639	0.7980 ± 0.0749
	F1	0.6306 ± 0.0393	0.4117 ± 0.0586	0.6741 ± 0.0496	0.7564 ± 0.0332	0.7508 ± 0.0811	0.2682 ± 0.1646	0.5469 ± 0.0403	0.7862 ± 0.0839	0.4159 ± 0.0698	0.8003 ± 0.0701
	NMI	0.5660 ± 0.0104	0.2357 ± 0.0481	0.5487 ± 0.0473	0.7821 ± 0.0321	0.6562 ± 0.0772	0.3054 ± 0.2115	0.3843 ± 0.0143	0.7407 ± 0.0313	0.2528 ± 0.0802	0.6872 ± 0.0892
	ARI	0.4231 ± 0.0204	0.1602 ± 0.0466	0.4382 ± 0.0499	0.7051 ± 0.0415	0.5669 ± 0.0959	0.1897 ± 0.1599	0.2816 ± 0.0319	0.6467 ± 0.0757	0.1578 ± 0.0647	0.6168 ± 0.1270
EthanolConcentration	ACC	0.2992 ± 0.0025	0.2786 ± 0.0051	0.3605 ± 0.0224	0.2634 ± 0.0096	0.2901 ± 0.0017	0.2962 ± 0.0172	0.2920 ± 0.0100	0.2962 ± 0.0044	0.2691 ± 0.0034	0.2969 ± 0.0153
	F1	0.2759 ± 0.0027	0.2529 ± 0.0093	0.3458 ± 0.0279	0.4681 ± 0.0189	0.2744 ± 0.0087	0.2442 ± 0.0306	0.2882 ± 0.0110	0.2747 ± 0.0067	0.2090 ± 0.0059	0.2732 ± 0.0102
	NMI	0.0149 ± 0.0009	0.0047 ± 0.0010	0.0574 ± 0.0189	0.0124 ± 0.0098	0.0116 ± 0.0018	0.0176 ± 0.0112	0.0074 ± 0.0029	0.0199 ± 0.0025	0.0025 ± 0.0015	0.0190 ± 0.0157
	ARI	0.0012 ± 0.0003	-0.0013 ± 0.0010	0.0307 ± 0.0121	0.0003 ± 0.0007	0.0032 ± 0.0005	0.0072 ± 0.0073	0.0008 ± 0.0024	0.0039 ± 0.0009	-0.0014 ± 0.0010	0.0097 ± 0.0119
FaceDetection	ACC	0.5130 ± 0.0030	OOM	0.5146 ± 0.0047	0.5051 ± 0.0029	0.5077 ± 0.0031	OOM	0.5038 ± 0.0018	0.5001 ± 0.0000	0.5054 ± 0.0040	0.5198 ± 0.0044
	F1	0.5125 ± 0.0028	OOM	0.4808 ± 0.0369	0.6113 ± 0.0585	0.5051 ± 0.0039	OOM	0.5028 ± 0.0023	0.3336 ± 0.0000	0.5030 ± 0.0060	0.5192 ± 0.0046
	NMI	0.0005 ± 0.0002	OOM	0.0010 ± 0.0004	0.0002 ± 0.0002	0.0002 ± 0.0001	OOM	0.0001 ± 0.0000	0.0027 ± 0.0000	0.0001 ± 0.0002	0.0012 ± 0.0005
	ARI	0.0006 ± 0.0003	OOM	0.0007 ± 0.0006	0.0001 ± 0.0001	0.0002 ± 0.0002	OOM	-0.0000 ± 0.0000	0.0000 ± 0.0000	0.0001 ± 0.0003	0.0015 ± 0.0006
FingerMovements	ACC	0.5250 ± 0.0052	0.5183 ± 0.0085	0.5680 ± 0.0160	0.5072 ± 0.0055	0.5163 ± 0.0127	0.5260 ± 0.0041	0.5264 ± 0.0026	0.5192 ± 0.0082	0.5130 ± 0.0054	0.5221 ± 0.0129
	F1	0.4951 ± 0.0111	0.4982 ± 0.0212	0.5349 ± 0.0218	0.6004 ± 0.0599	0.5051 ± 0.0184	0.5230 ± 0.0023	0.4997 ± 0.0023	0.5161 ± 0.0058	0.5124 ± 0.0051	0.5183 ± 0.0074
	NMI	0.0027 ± 0.0009	0.0014 ± 0.0009	0.0210 ± 0.0132	0.0023 ± 0.0035	0.0013 ± 0.0016	0.0021 ± 0.0007	0.0028 ± 0.0005	0.0013 ± 0.0009	0.0006 ± 0.0003	0.0022 ± 0.0026
	ARI	0.0008 ± 0.0009	-0.0004 ± 0.0009	0.0121 ± 0.0087	-0.0010 ± 0.0008	-0.0005 ± 0.0021	0.0004 ± 0.0009	0.0009 ± 0.0005	-0.0006 ± 0.0013	-0.0016 ± 0.0004	0.0003 ± 0.0031
HandMovementDirection	ACC	0.2829 ± 0.0063	0.2957 ± 0.0087	0.4405 ± 0.0359	0.3248 ± 0.0164	0.3017 ± 0.0088	0.3017 ± 0.0123	0.3026 ± 0.0099	0.3120 ± 0.0097	0.3743 ± 0.0375	0.3179 ± 0.0100
	F1	0.2108 ± 0.0397	0.2907 ± 0.0061	0.4070 ± 0.0226	0.3328 ± 0.0189	0.2867 ± 0.0112	0.2667 ± 0.0205	0.2996 ± 0.0128	0.3089 ± 0.0099	0.3667 ± 0.0383	0.3145 ± 0.0104
	NMI	0.0168 ± 0.0104	0.0156 ± 0.0061	0.0910 ± 0.0245	0.0286 ± 0.0110	0.0184 ± 0.0049	0.0206 ± 0.0103	0.0144 ± 0.0047	0.0208 ± 0.0045	0.0818 ± 0.0394	0.0208 ± 0.0050
	ARI	-0.0047 ± 0.0038	0.0015 ± 0.0062	0.0475 ± 0.0166	0.0109 ± 0.0104	0.0001 ± 0.0045	0.0007 ± 0.0032	0.0003 ± 0.0032	0.0046 ± 0.0035	0.0488 ± 0.0282	0.0052 ± 0.0047
Handwriting	ACC	0.2433 ± 0.0041	0.1264 ± 0.0027	0.1536 ± 0.0097	0.1090 ± 0.0209	0.1778 ± 0.0097	0.0878 ± 0.0160	0.1708 ± 0.0101	0.1488 ± 0.0067	0.2940 ± 0.0068	0.2530 ± 0.0126
	F1	0.2418 ± 0.0080	0.1260 ± 0.0020	0.1501 ± 0.0114	0.1206 ± 0.0171	0.1741 ± 0.0071	0.0352 ± 0.0127	0.1679 ± 0.0098	0.1484 ± 0.0078	0.2881 ± 0.0075	0.2591 ± 0.0132
	NMI	0.3723 ± 0.0061	0.1523 ± 0.0052	0.1934 ± 0.0070	0.1228 ± 0.0510	0.2497 ± 0.0235	0.0637 ± 0.0274	0.2260 ± 0.0097	0.2009 ± 0.0035	0.0101 ± 0.0049	0.3652 ± 0.0139
	ARI	0.1106 ± 0.0035	0.0125 ± 0.0016	0.0254 ± 0.0036	0.0195 ± 0.0090	0.0494 ± 0.0107	0.0084 ± 0.0048	0.0429 ± 0.0045	0.0282 ± 0.0012	-0.0047 ± 0.0030	0.1134 ± 0.0113

spectral structure, or weak label-to-signal correspondence. For example, FCACC, TFMCC, EMTC, FEI, and k-Graph obtain the best result on some individual datasets. Thus, the key observation is not dominance on every dataset, but stable average behavior across different sample sizes, sequence lengths, variable dimensions, and label-provided cluster numbers k. Fig. 1 provides the corresponding Bonferroni-Dunn test. Average ranks are computed over 30 datasets and four metrics, and lower ranks indicate stronger overall clustering quality. The critical-difference bars show which rank gaps are statistically distinguishable at $\alpha = 0.1$ and $\alpha = 0.05$, giving a statistical view of the cross-dataset consistency reported in the tables.

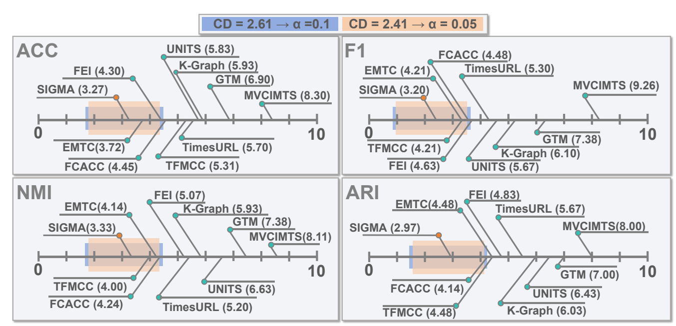

Fig. 1. Bonferroni-Dunn rank test on 30 UEA datasets under ACC, F1, NMI, and ARI. Lower average rank indicates stronger clustering quality, and critical-difference bars report significance thresholds at $\alpha = 0.1$ and $\alpha = 0.05$.

B. Complete Module-Level and Route-Level Ablations

The main paper reports averaged ablations for Q2, and this supplemental material subsection expands the same analysis to all 30 UEA datasets. All variants follow the protocol in Section II. The module-level ablations remove CCMA and MEVG separately or jointly, testing how patch participation

TABLE II

COMPLETE CLUSTERING RESULTS ON 30 UEA DATASETS EVALUATED BY ACC, F1, NMI, AND ARI, PART II. THE **BEST** AND **SECOND BEST** RESULTS ARE HIGHLIGHTED BY CELL BACKGROUND.

Datasets	Metric	FCACC (PR'26)	GTM (ICLR'26)	EMTC (AAAI'26)	TFMCC (AAAI'26)	FEI (AAAI'25)	MVCIMTS (INFFUS'25)	*k*-Graph (TKDE'25)	TimesURL (AAAI'24)	UNITS (NeurIPS'24)	SIGMA (ours)
Heartbeat	ACC	0.5844 ± 0.0079	0.5340 ± 0.0292	0.7180 ± 0.0020	0.7076 ± 0.0476	0.7090 ± 0.0000	0.6758 ± 0.0497	0.7296 ± 0.0025	0.7149 ± 0.0106	0.6244 ± 0.0238	0.6817 ± 0.0546
	F1	0.5688 ± 0.0055	0.4974 ± 0.0193	0.7553 ± 0.1598	0.6934 ± 0.0669	0.8298 ± 0.0000	0.4592 ± 0.0483	0.5461 ± 0.0071	0.7922 ± 0.0760	0.5347 ± 0.0192	0.5944 ± 0.0367
	NMI	0.0389 ± 0.0016	0.0012 ± 0.0015	0.0106 ± 0.0023	0.0465 ± 0.0073	0.0204 ± 0.0000	0.0016 ± 0.0020	0.0344 ± 0.0041	0.0300 ± 0.0207	0.0070 ± 0.0077	0.0382 ± 0.0220
	ARI	0.0258 ± 0.0062	0.0030 ± 0.0062	-0.0029 ± 0.0061	0.0966 ± 0.0306	-0.0144 ± 0.0000	0.0018 ± 0.0101	0.0875 ± 0.0071	0.0220 ± 0.0714	0.0243 ± 0.0111	0.0981 ± 0.0485
InsectWingbeat	ACC	0.1651 ± 0.0020	0.1003 ± 0.0000	0.1251 ± 0.0049	0.1099 ± 0.0024	0.1533 ± 0.0004	OOM	0.1451 ± 0.0010	0.1000 ± 0.0000	0.0872 ± 0.0071	0.1336 ± 0.0017
	F1	0.1623 ± 0.0020	0.0189 ± 0.0001	0.0945 ± 0.0222	0.0962 ± 0.0074	0.1062 ± 0.0051	OOM	0.1374 ± 0.0049	0.1818 ± 0.0000	0.0812 ± 0.0055	0.1209 ± 0.0043
	NMI	0.0428 ± 0.0011	0.0010 ± 0.0001	0.0090 ± 0.0015	0.1883 ± 0.0060	0.0339 ± 0.0008	OOM	0.0210 ± 0.0008	0.0000 ± 0.0000	0.0032 ± 0.0061	0.0146 ± 0.0015
	ARI	0.0216 ± 0.0005	0.0000 ± 0.0000	0.0027 ± 0.0013	0.0377 ± 0.0055	0.0130 ± 0.0007	OOM	0.0087 ± 0.0002	0.0000 ± 0.0000	0.0043 ± 0.0000	0.0059 ± 0.0006
JapaneseVowels	ACC	0.6286 ± 0.0530	0.1844 ± 0.0000	0.3203 ± 0.0273	0.4766 ± 0.0639	0.5198 ± 0.0223	0.1844 ± 0.0000	0.3369 ± 0.0199	0.1844 ± 0.0000	0.2619 ± 0.0271	0.8525 ± 0.0431
	F1	0.6617 ± 0.0303	0.0346 ± 0.0000	0.3325 ± 0.0365	0.4274 ± 0.0355	0.5551 ± 0.0053	0.0346 ± 0.0000	0.3069 ± 0.0236	0.3113 ± 0.0000	0.2755 ± 0.0282	0.8494 ± 0.0383
	NMI	0.5950 ± 0.0134	0.0000 ± 0.0000	0.2410 ± 0.0418	0.5250 ± 0.0397	0.4727 ± 0.0084	0.0000 ± 0.0000	0.2820 ± 0.0112	0.0000 ± 0.0000	0.1643 ± 0.0209	0.8113 ± 0.0180
	ARI	0.4771 ± 0.0389	0.0000 ± 0.0000	0.1269 ± 0.0412	0.3167 ± 0.0466	0.3049 ± 0.0281	0.0000 ± 0.0000	0.1511 ± 0.0210	0.0000 ± 0.0000	0.0648 ± 0.0171	0.7578 ± 0.0423
Libras	ACC	0.3822 ± 0.0052	0.3894 ± 0.0321	0.4011 ± 0.0315	0.3544 ± 0.0099	0.4317 ± 0.0256	0.1456 ± 0.0416	0.3844 ± 0.0208	0.2700 ± 0.0148	0.2839 ± 0.0116	0.3711 ± 0.0125
	F1	0.3931 ± 0.0193	0.3772 ± 0.0369	0.4028 ± 0.0406	0.2626 ± 0.0226	0.4438 ± 0.0298	0.0627 ± 0.0302	0.3700 ± 0.0242	0.2767 ± 0.0166	0.2853 ± 0.0157	0.3655 ± 0.0158
	NMI	0.5311 ± 0.0095	0.4697 ± 0.0327	0.4890 ± 0.0322	0.4698 ± 0.0245	0.5056 ± 0.0189	0.1311 ± 0.0783	0.5126 ± 0.0062	0.3101 ± 0.0185	0.3562 ± 0.0152	0.4806 ± 0.0259
	ARI	0.2484 ± 0.0082	0.2143 ± 0.0276	0.1881 ± 0.0295	0.2019 ± 0.0228	0.2423 ± 0.0142	0.0370 ± 0.0255	0.2532 ± 0.0127	0.0901 ± 0.0166	0.1246 ± 0.0146	0.2089 ± 0.0196
LSST	ACC	0.3434 ± 0.0022	0.1224 ± 0.0038	0.3672 ± 0.0074	0.2491 ± 0.0169	0.3327 ± 0.0037	0.3227 ± 0.0092	0.1578 ± 0.0113	0.3293 ± 0.0025	0.1805 ± 0.0098	0.2019 ± 0.0029
	F1	0.1918 ± 0.0134	0.0984 ± 0.0072	0.2055 ± 0.0297	0.2114 ± 0.0148	0.1689 ± 0.0141	0.0521 ± 0.0222	0.1149 ± 0.0053	0.2387 ± 0.0157	0.1376 ± 0.0084	0.1840 ± 0.0147
	NMI	0.1188 ± 0.0020	0.0311 ± 0.0056	0.1959 ± 0.0261	0.2066 ± 0.0198	0.1338 ± 0.0040	0.0237 ± 0.0303	0.0764 ± 0.0061	0.1398 ± 0.0038	0.0543 ± 0.0132	0.1396 ± 0.0043
	ARI	0.0453 ± 0.0007	0.0096 ± 0.0022	0.0936 ± 0.0182	0.0634 ± 0.0311	0.0859 ± 0.0106	0.0070 ± 0.0089	0.0222 ± 0.0040	0.0254 ± 0.0061	0.0207 ± 0.0068	0.0549 ± 0.0028
MotorImagery	ACC	0.5016 ± 0.0013	0.5360 ± 0.0147	0.5840 ± 0.0258	0.5048 ± 0.0072	0.5058 ± 0.0035	0.5111 ± 0.0059	0.5159 ± 0.0111	0.5127 ± 0.0048	0.5333 ± 0.0173	0.5508 ± 0.0070
	F1	0.3512 ± 0.0022	0.5084 ± 0.0319	0.5308 ± 0.0377	0.6586 ± 0.0706	0.3794 ± 0.0205	0.4565 ± 0.0416	0.5093 ± 0.0107	0.4401 ± 0.0338	0.5180 ± 0.0271	0.5362 ± 0.0175
	NMI	0.0003 ± 0.0002	0.0055 ± 0.0031	0.0540 ± 0.0291	0.0023 ± 0.0040	0.0011 ± 0.0008	0.0017 ± 0.0016	0.0012 ± 0.0011	0.0012 ± 0.0004	0.0044 ± 0.0042	0.0096 ± 0.0034
	ARI	-0.0002 ± 0.0000	0.0040 ± 0.0035	0.0254 ± 0.0160	-0.0003 ± 0.0004	-0.0003 ± 0.0001	-0.0010 ± 0.0011	-0.0010 ± 0.0014	-0.0006 ± 0.0001	0.0033 ± 0.0059	0.0082 ± 0.0028
NATOPS	ACC	0.6537 ± 0.0975	0.4861 ± 0.0512	0.3978 ± 0.0514	0.5672 ± 0.0859	0.4350 ± 0.0268	0.3267 ± 0.0919	0.3817 ± 0.0194	0.3522 ± 0.0811	0.2972 ± 0.0249	0.5022 ± 0.0157
	F1	0.6529 ± 0.0968	0.4643 ± 0.0539	0.4342 ± 0.1010	0.5167 ± 0.0816	0.4372 ± 0.0599	0.2052 ± 0.0868	0.3871 ± 0.0168	0.2131 ± 0.0691	0.2951 ± 0.0249	0.4958 ± 0.0103
	NMI	0.6454 ± 0.0805	0.4355 ± 0.0551	0.2990 ± 0.1190	0.5575 ± 0.0724	0.3896 ± 0.0459	0.2876 ± 0.1673	0.1897 ± 0.0207	0.4115 ± 0.1613	0.1097 ± 0.0324	0.5159 ± 0.0230
	ARI	0.5091 ± 0.1162	0.3056 ± 0.0590	0.1759 ± 0.0826	0.4179 ± 0.0983	0.2563 ± 0.0459	0.1603 ± 0.1191	0.1147 ± 0.0214	0.2551 ± 0.1261	0.0549 ± 0.0222	0.3489 ± 0.0284
PEMS-SF	ACC	0.4864 ± 0.0558	0.1968 ± 0.0065	0.4301 ± 0.0842	0.2268 ± 0.0179	0.3582 ± 0.0366	0.2514 ± 0.0897	0.4268 ± 0.0172	0.2377 ± 0.0102	0.4409 ± 0.0466	0.4436 ± 0.0844
	F1	0.5110 ± 0.0447	0.1941 ± 0.0064	0.4202 ± 0.0807	0.2263 ± 0.0204	0.3782 ± 0.0333	0.1185 ± 0.0734	0.4239 ± 0.0214	0.2370 ± 0.0154	0.4683 ± 0.0591	0.4902 ± 0.1198
	NMI	0.4510 ± 0.0416	0.0348 ± 0.0119	0.3687 ± 0.1015	0.1270 ± 0.0168	0.3336 ± 0.0281	0.2715 ± 0.2226	0.4593 ± 0.0196	0.1742 ± 0.0226	0.4167 ± 0.0441	0.4570 ± 0.1104
	ARI	0.2911 ± 0.0478	0.0030 ± 0.0034	0.2134 ± 0.0924	0.0374 ± 0.0156	0.1673 ± 0.0312	0.1333 ± 0.1105	0.2557 ± 0.0121	0.0651 ± 0.0082	0.2471 ± 0.0435	0.2839 ± 0.0947
PenDigits	ACC	0.4246 ± 0.0409	0.4128 ± 0.0672	0.4089 ± 0.0499	0.6548 ± 0.0159	0.3484 ± 0.0354	0.2670 ± 0.0505	0.4078 ± 0.0162	0.3647 ± 0.0150	0.4687 ± 0.0517	0.6841 ± 0.0284
	F1	0.4116 ± 0.0303	0.3971 ± 0.0746	0.4086 ± 0.0578	0.5693 ± 0.0171	0.3548 ± 0.0321	0.1710 ± 0.0688	0.3825 ± 0.0213	0.3747 ± 0.0150	0.4542 ± 0.0512	0.6938 ± 0.0411
	NMI	0.3233 ± 0.0199	0.3405 ± 0.0616	0.3994 ± 0.0617	0.6583 ± 0.0086	0.3059 ± 0.0165	0.2076 ± 0.0559	0.3390 ± 0.0131	0.2611 ± 0.0165	0.4348 ± 0.0630	0.6485 ± 0.0189
	ARI	0.2218 ± 0.0289	0.2151 ± 0.0560	0.2313 ± 0.0515	0.5196 ± 0.0193	0.1503 ± 0.0122	0.1033 ± 0.0425	0.2183 ± 0.0158	0.1528 ± 0.0112	0.2822 ± 0.0658	0.5305 ± 0.0276
PhonemeSpectra	ACC	0.1149 ± 0.0003	0.0633 ± 0.0006	0.0963 ± 0.0032	0.1124 ± 0.0038	0.1283 ± 0.0024	0.0454 ± 0.0059	0.0916 ± 0.0028	0.0871 ± 0.0023	0.0812 ± 0.0071	0.1127 ± 0.0015
	F1	0.1159 ± 0.0004	0.0605 ± 0.0010	0.0944 ± 0.0004	0.1048 ± 0.0067	0.1242 ± 0.0038	0.0099 ± 0.0028	0.0913 ± 0.0033	0.0845 ± 0.0017	0.0786 ± 0.0080	0.1126 ± 0.0017
	NMI	0.1793 ± 0.0023	0.0451 ± 0.0015	0.1386 ± 0.0089	0.2039 ± 0.0025	0.2085 ± 0.0038	0.0366 ± 0.0136	0.1090 ± 0.0036	0.1174 ± 0.0070	0.0932 ± 0.0191	0.1758 ± 0.0026
	ARI	0.0339 ± 0.0007	0.0038 ± 0.0003	0.0166 ± 0.0040	0.0424 ± 0.0048	0.0449 ± 0.0011	0.0046 ± 0.0026	0.0178 ± 0.0015	0.0186 ± 0.0020	0.0143 ± 0.0045	0.0334 ± 0.0008
RacketSports	ACC	0.4172 ± 0.0058	0.3921 ± 0.0150	0.4355 ± 0.0286	0.6205 ± 0.0400	0.4257 ± 0.0251	0.3129 ± 0.0296	0.3551 ± 0.0324	0.5036 ± 0.0475	0.3743 ± 0.0375	0.6007 ± 0.0721
	F1	0.4138 ± 0.0065	0.3898 ± 0.0135	0.4064 ± 0.0374	0.5330 ± 0.0327	0.4067 ± 0.0328	0.2105 ± 0.0508	0.3529 ± 0.0340	0.5273 ± 0.0345	0.3667 ± 0.0383	0.6006 ± 0.0707
	NMI	0.1398 ± 0.0042	0.1024 ± 0.0386	0.1138 ± 0.0264	0.4687 ± 0.0339	0.1250 ± 0.0289	0.0688 ± 0.0371	0.0601 ± 0.0348	0.3279 ± 0.0420	0.0818 ± 0.0394	0.3488 ± 0.0845
	ARI	0.0880 ± 0.0042	0.0590 ± 0.0167	0.0776 ± 0.0307	0.3723 ± 0.0445	0.0924 ± 0.0200	0.0127 ± 0.0191	0.0400 ± 0.0293	0.2394 ± 0.0441	0.0488 ± 0.0282	0.2858 ± 0.0965
SelfRegulationSCP1	ACC	0.5362 ± 0.0031	0.5447 ± 0.0307	0.7126 ± 0.0431	0.5027 ± 0.0000	0.6096 ± 0.0472	0.5537 ± 0.0342	0.5586 ± 0.0087	0.7340 ± 0.0059	0.5811 ± 0.0402	0.7754 ± 0.0128
	F1	0.5081 ± 0.0040	0.5434 ± 0.0324	0.7073 ± 0.0419	0.7065 ± 0.0000	0.6082 ± 0.0487	0.4723 ± 0.0964	0.5584 ± 0.0087	0.7184 ± 0.0076	0.5676 ± 0.0372	0.7745 ± 0.0125
	NMI	0.0059 ± 0.0009	0.0086 ± 0.0093	0.1642 ± 0.0748	0.0000 ± 0.0000	0.0421 ± 0.0290	0.0261 ± 0.0139	0.0102 ± 0.0031	0.2493 ± 0.0044	0.0328 ± 0.0305	0.2382 ± 0.0246
	ARI	0.0039 ± 0.0009	0.0100 ± 0.0128	0.1857 ± 0.0753	0.0000 ± 0.0000	0.0553 ± 0.0393	0.0154 ± 0.0124	0.0123 ± 0.0043	0.2182 ± 0.0111	0.0313 ± 0.0296	0.3028 ± 0.0274
SelfRegulationSCP2	ACC	0.5163 ± 0.0051	0.5258 ± 0.0162	0.5811 ± 0.0143	0.5027 ± 0.0000	0.5521 ± 0.0096	0.5347 ± 0.0210	0.5053 ± 0.0069	0.5432 ± 0.0054	0.5811 ± 0.0402	0.5253 ± 0.0141
	F1	0.4913 ± 0.0063	0.5242 ± 0.0166	0.5657 ± 0.0301	0.7065 ± 0.0000	0.5448 ± 0.0077	0.5302 ± 0.0239	0.5039 ± 0.0073	0.5113 ± 0.0110	0.5676 ± 0.0372	0.5163 ± 0.0084
	NMI	0.0011 ± 0.0006	0.0027 ± 0.0027	0.0234 ± 0.0043	0.0000 ± 0.0000	0.0090 ± 0.0034	0.0049 ± 0.0048	0.0002 ± 0.0004	0.0083 ± 0.0020	0.0328 ± 0.0305	0.0034 ± 0.0044
	ARI	-0.0010 ± 0.0006	0.0011 ± 0.0036	0.0224 ± 0.0083	0.0000 ± 0.0000	0.0088 ± 0.0040	0.0041 ± 0.0065	-0.0023 ± 0.0005	0.0056 ± 0.0018	0.0313 ± 0.0296	0.0009 ± 0.0038
SpokenArabicDigits	ACC	0.4902 ± 0.0258	0.1000 ± 0.0000	0.2415 ± 0.0313	0.3869 ± 0.0543	0.4655 ± 0.1073	0.1889 ± 0.0452	0.3692 ± 0.0053	0.1000 ± 0.0000	0.3054 ± 0.0040	0.4420 ± 0.0257
	F1	0.4939 ± 0.0288	0.0182 ± 0.0000	0.2320 ± 0.0343	0.3303 ± 0.0362	0.4617 ± 0.1044	0.1196 ± 0.0519	0.3713 ± 0.0084	0.1819 ± 0.0000	0.3030 ± 0.0060	0.4460 ± 0.0206
	NMI	0.4740 ± 0.0129	0.0000 ± 0.0000	0.1566 ± 0.0474	0.4331 ± 0.0779	0.3946 ± 0.0887	0.1032 ± 0.0526	0.2691 ± 0.0144	0.0000 ± 0.0000	0.2431 ± 0.0002	0.4093 ± 0.0258
	ARI	0.3368 ± 0.0189	0.0000 ± 0.0000	0.0698 ± 0.0246	0.1351 ± 0.0703	0.2759 ± 0.0945	0.0529 ± 0.0279	0.1723 ± 0.0074	0.0000 ± 0.0000	0.1591 ± 0.0013	0.2643 ± 0.0280
StandWalkJump	ACC	0.4741 ± 0.0148	0.4593 ± 0.0502	0.6133 ± 0.0499	0.4444 ± 0.0406	0.5259 ± 0.0718	0.3333 ± 0.0000	0.4370 ± 0.0363	0.3926 ± 0.0181	0.5407 ± 0.0554	0.4296 ± 0.0646
	F1	0.4660 ± 0.0192	0.4545 ± 0.0527	0.6319 ± 0.0356	0.4183 ± 0.0631	0.5214 ± 0.0775	0.1667 ± 0.0000	0.4344 ± 0.0405	0.4802 ± 0.0242	0.5200 ± 0.0589	0.4069 ± 0.0675
	NMI	0.0997 ± 0.0235	0.1106 ± 0.0834	0.3446 ± 0.0358	0.1139 ± 0.0421	0.1219 ± 0.0596	0.0000 ± 0.0000	0.0866 ± 0.0489	0.0724 ± 0.0121	0.2150 ± 0.1475	0.0664 ± 0.0435
	ARI	0.0337 ± 0.0237	0.0184 ± 0.0621	0.1583 ± 0.0329	0.0137 ± 0.0393	0.0373 ± 0.0581	0.0000 ± 0.0000	-0.0145 ± 0.0314	-0.0146 ± 0.0145	0.1035 ± 0.1257	0.0083 ± 0.0492
UWaveGestureLibrary	ACC	0.5447 ± 0.0363	0.3073 ± 0.0291	0.3844 ± 0.0327	0.4614 ± 0.0993	0.3664 ± 0.0330	0.1250 ± 0.0000	0.3295 ± 0.0220	0.4559 ± 0.0421	0.5211 ± 0.0154	0.4550 ± 0.0410
	F1	0.5371 ± 0.0370	0.2952 ± 0.0280	0.3543 ± 0.0294	0.3472 ± 0.0630	0.3633 ± 0.0246	0.0278 ± 0.0000	0.3318 ± 0.0210	0.4536 ± 0.0487	0.5130 ± 0.0167	0.4412 ± 0.0434
	NMI	0.4631 ± 0.0409	0.1760 ± 0.0179	0.2606 ± 0.0468	0.3891 ± 0.0861	0.2394 ± 0.0115	0.0000 ± 0.0000	0.1658 ± 0.0284	0.3457 ± 0.0256	0.0022 ± 0.0026	0.3311 ± 0.0470
	ARI	0.3593 ± 0.0466	0.0901 ± 0.0148	0.1662 ± 0.0356	0.1692 ± 0.1171	0.1403 ± 0.0195	0.0000 ± 0.0000	0.0960 ± 0.0250	0.2268 ± 0.0340	0.0003 ± 0.0032	0.2016 ± 0.0428

is selected before encoding and how admitted content is organized before routed optimization. The route-level ablations remove reconstruction and contrastive learning separately or jointly, testing how retained views are anchored for temporal recovery and refined toward prototype-aware grouping.

TABLE III and TABLE IV report the complete module-level results. The full SIGMA model gives the strongest average behavior, although individual datasets may favor a simpler variant on a single metric. Such variation is expected on UEA because similarity cues differ across motion, speech, sensor, and biomedical records. The main pattern lies in the failure caused by each removal. Without CCMA, background-dominated patches can enter encoder-side interaction before representation learning becomes selective. Without MEVG, admitted patches lose complementary view organization, making the retained representation depend more on narrow local descriptions. Removing both modules usually leads to the clearest degradation, indicating that source-side selection and endogenous-view compensation play complementary roles.

TABLE V and TABLE VI further report the route-level results. Removing the reconstruction route weakens the temporal-content anchor after masking, so the learned representation becomes more sensitive to partial observation. Removing the contrastive route leaves retained semantics insufficiently aligned with prototype-level structure, which can yield plausible embeddings but weak partitions. Joint route removal gives the most evident collapse, especially on ArticularyWordRecognition, CharacterTrajectories, JapaneseVowels, and InsectWingbeat, where NMI and ARI drop sharply. These failures show that admitted content alone is not enough for

TABLE III
ABLATION STUDIES ON MODULE DESIGNS OF SIGMA, PART I. RED "‡" MARKS DEGRADATION FROM COMPLETE SIGMA.

Datasets (1-15)	Components		Metrics			
	CCMA	MEVG	ACC	F1	NMI	ARI
ArticularyWord Recognition	✓	✓	0.7329 ± 0.0395	0.7696 ± 0.0443	0.8037 ± 0.0246	0.6029 ± 0.0470
		✓	0.7155 ± 0.0137‡	0.7451 ± 0.0215‡	0.7824 ± 0.0132‡	0.5728 ± 0.0248‡
	✓		0.7739 ± 0.0437	0.7997 ± 0.0481	0.8255 ± 0.0189	0.6570 ± 0.0427
			0.6918 ± 0.0306‡	0.7336 ± 0.0245‡	0.7709 ± 0.0187‡	0.5517 ± 0.0369‡
AtrialFibrillation	✓	✓	0.4933 ± 0.0327	0.4720 ± 0.0232	0.1215 ± 0.0474	0.0443 ± 0.0492
		✓	0.4867 ± 0.0163‡	0.4331 ± 0.0116‡	0.1050 ± 0.0047‡	0.0404 ± 0.0051‡
	✓		0.4800 ± 0.0400‡	0.4681 ± 0.0854‡	0.1365 ± 0.0448	0.0664 ± 0.0421
			0.4600 ± 0.0327‡	0.5024 ± 0.0485	0.1000 ± 0.0087‡	0.0346 ± 0.0071‡
BasicMotions	✓	✓	0.9675 ± 0.0150	0.9671 ± 0.0154	0.9273 ± 0.0277	0.9169 ± 0.0363
		✓	0.9825 ± 0.0127	0.9824 ± 0.0129	0.9497 ± 0.0366	0.9533 ± 0.0338
	✓		0.8900 ± 0.1317‡	0.9203 ± 0.0746‡	0.8924 ± 0.0754‡	0.8361 ± 0.1360‡
			0.9700 ± 0.0100	0.9698 ± 0.0102	0.9138 ± 0.0287‡	0.9202 ± 0.0263
Character Trajectories	✓	✓	0.7879 ± 0.0499	0.8091 ± 0.0408	0.8463 ± 0.0284	0.7214 ± 0.0576
		✓	0.7824 ± 0.0547‡	0.8177 ± 0.0436	0.8416 ± 0.0367‡	0.7078 ± 0.0644‡
	✓		0.7371 ± 0.0227‡	0.7610 ± 0.0346‡	0.8038 ± 0.0198‡	0.6417 ± 0.0303‡
			0.7369 ± 0.0590‡	0.7699 ± 0.0708‡	0.8138 ± 0.0468‡	0.6556 ± 0.0750‡
Cricket	✓	✓	0.8244 ± 0.0255	0.8463 ± 0.0115	0.8522 ± 0.0167	0.7259 ± 0.0299
		✓	0.8489 ± 0.0349	0.8739 ± 0.0420	0.8737 ± 0.0444	0.7589 ± 0.0659
	✓		0.7811 ± 0.0658‡	0.8191 ± 0.0588‡	0.8346 ± 0.0364‡	0.6817 ± 0.0702‡
			0.8078 ± 0.0114‡	0.8292 ± 0.0241‡	0.8376 ± 0.0130‡	0.6937 ± 0.0195‡
DuckDuckGeese	✓	✓	0.3900 ± 0.0141	0.3730 ± 0.0193	0.1749 ± 0.0375	0.0795 ± 0.0345
		✓	0.3700 ± 0.0179‡	0.3600 ± 0.0253‡	0.1143 ± 0.0234‡	0.0526 ± 0.0108‡
	✓		0.3280 ± 0.0421‡	0.2966 ± 0.0425‡	0.0869 ± 0.0246‡	0.0196 ± 0.0205‡
			0.3680 ± 0.0194‡	0.3449 ± 0.0235‡	0.1229 ± 0.0145‡	0.0306 ± 0.0112‡
ERing	✓	✓	0.7307 ± 0.0629	0.7287 ± 0.0680	0.5996 ± 0.0590	0.5124 ± 0.0760
		✓	0.6867 ± 0.0669‡	0.7380 ± 0.0341	0.6238 ± 0.0426	0.5008 ± 0.0669‡
	✓		0.7300 ± 0.0485‡	0.7313 ± 0.0475	0.6198 ± 0.0400	0.5226 ± 0.0569
			0.7173 ± 0.0259‡	0.7082 ± 0.0253‡	0.6112 ± 0.0390	0.5033 ± 0.0415‡
EigenWorms	✓	✓	0.4193 ± 0.0788	0.3646 ± 0.0438	0.1612 ± 0.0594	0.1515 ± 0.0726
		✓	0.4131 ± 0.0101‡	0.3546 ± 0.0347‡	0.2196 ± 0.0211	0.1907 ± 0.0156
	✓		0.4425 ± 0.0451	0.4077 ± 0.0359	0.1829 ± 0.0256	0.1469 ± 0.0344‡
			0.3822 ± 0.0462‡	0.3397 ± 0.0392‡	0.1713 ± 0.0600	0.1368 ± 0.0716‡
Epilepsy	✓	✓	0.4655 ± 0.0229	0.4573 ± 0.0233	0.2355 ± 0.0362	0.1668 ± 0.0336
		✓	0.4640 ± 0.0291‡	0.4648 ± 0.0300	0.2217 ± 0.0332‡	0.1511 ± 0.0259‡
	✓		0.4720 ± 0.0368	0.4705 ± 0.0382	0.2330 ± 0.0451‡	0.1551 ± 0.0279‡
			0.4829 ± 0.0238	0.4847 ± 0.0243	0.2418 ± 0.0298	0.1637 ± 0.0212‡
Ethanol Concentration	✓	✓	0.3061 ± 0.0208	0.2766 ± 0.0066	0.0199 ± 0.0156	0.0108 ± 0.0123
		✓	0.3015 ± 0.0099‡	0.2777 ± 0.0082	0.0114 ± 0.0033‡	0.0049 ± 0.0030‡
	✓		0.3088 ± 0.0183	0.2881 ± 0.0252	0.0184 ± 0.0066‡	0.0104 ± 0.0062‡
			0.2950 ± 0.0092‡	0.2797 ± 0.0108	0.0124 ± 0.0061‡	0.0045 ± 0.0039‡
FaceDetection	✓	✓	0.5107 ± 0.0042	0.5100 ± 0.0049	0.0004 ± 0.0003	0.0004 ± 0.0004
		✓	0.5051 ± 0.0028‡	0.5050 ± 0.0028‡	0.0001 ± 0.0001‡	0.0000 ± 0.0001‡
	✓		0.5101 ± 0.0023‡	0.5092 ± 0.0027‡	0.0003 ± 0.0001‡	0.0003 ± 0.0002‡
			0.5063 ± 0.0076‡	0.5061 ± 0.0077‡	0.0003 ± 0.0005‡	0.0003 ± 0.0007‡
FingerMovements	✓	✓	0.5188 ± 0.0085	0.5142 ± 0.0086	0.0013 ± 0.0007	0.0006 ± 0.0010
		✓	0.5168 ± 0.0101‡	0.5123 ± 0.0056‡	0.0013 ± 0.0015‡	0.0008 ± 0.0017
	✓		0.5168 ± 0.0128‡	0.5125 ± 0.0160‡	0.0013 ± 0.0014‡	0.0005 ± 0.0019‡
			0.5130 ± 0.0074‡	0.5127 ± 0.0073‡	0.0006 ± 0.0007‡	0.0015 ± 0.0010
HandMovement Direction	✓	✓	0.3333 ± 0.0143	0.3298 ± 0.0171	0.0205 ± 0.0063	0.0060 ± 0.0052
		✓	0.3051 ± 0.0154‡	0.3027 ± 0.0162‡	0.0139 ± 0.0036‡	0.0008 ± 0.0029‡
	✓		0.3179 ± 0.0249‡	0.3148 ± 0.0246‡	0.0201 ± 0.0119‡	0.0049 ± 0.0111‡
			0.3060 ± 0.0126‡	0.3034 ± 0.0116‡	0.0144 ± 0.0029‡	0.0011 ± 0.0025‡
Handwriting	✓	✓	0.2386 ± 0.0181	0.2433 ± 0.0195	0.3461 ± 0.0286	0.1006 ± 0.0180
		✓	0.2102 ± 0.0132‡	0.2103 ± 0.0137‡	0.3030 ± 0.0188‡	0.0770 ± 0.0111‡
	✓		0.2444 ± 0.0113	0.2460 ± 0.0156	0.3503 ± 0.0160	0.1050 ± 0.0109
			0.2224 ± 0.0130‡	0.2234 ± 0.0113‡	0.3212 ± 0.0209‡	0.0852 ± 0.0100‡
Heartbeat	✓	✓	0.6636 ± 0.0589	0.5679 ± 0.0507	0.0391 ± 0.0232	0.0689 ± 0.0404
		✓	0.6826 ± 0.0369	0.6002 ± 0.0271	0.0438 ± 0.0108	0.0952 ± 0.0277
	✓		0.6841 ± 0.0410	0.5474 ± 0.0747‡	0.0328 ± 0.0106‡	0.0620 ± 0.0591‡
			0.7066 ± 0.0091	0.6463 ± 0.0947	0.0365 ± 0.0108‡	0.0846 ± 0.0516

TABLE IV
ABLATION STUDIES ON MODULE DESIGNS OF SIGMA, PART II. RED "‡" MARKS DEGRADATION FROM COMPLETE SIGMA.

Datasets (16-30)	Components		Metrics			
	CCMA	MEVG	ACC	F1	NMI	ARI
InsectWingbeat	✓	✓	0.1344 ± 0.0023	0.1202 ± 0.0033	0.0153 ± 0.0012	0.0061 ± 0.0007
		✓	0.1351 ± 0.0015	0.1206 ± 0.0036	0.0147 ± 0.0008‡	0.0064 ± 0.0001
	✓		0.1304 ± 0.0032‡	0.1115 ± 0.0076‡	0.0132 ± 0.0011‡	0.0048 ± 0.0013‡
			0.1323 ± 0.0039‡	0.1099 ± 0.0069‡	0.0139 ± 0.0017‡	0.0050 ± 0.0011‡
JapaneseVowels	✓	✓	0.8647 ± 0.0447	0.8626 ± 0.0432	0.8195 ± 0.0260	0.7682 ± 0.0565
		✓	0.8162 ± 0.0393‡	0.8557 ± 0.0416‡	0.8044 ± 0.0126‡	0.7321 ± 0.0429‡
	✓		0.8750 ± 0.0347	0.8840 ± 0.0294	0.8220 ± 0.0239	0.7847 ± 0.0385
			0.8800 ± 0.0306	0.8802 ± 0.0270	0.8140 ± 0.0272‡	0.7648 ± 0.0553‡
LSST	✓	✓	0.2042 ± 0.0087	0.1805 ± 0.0063	0.1456 ± 0.0063	0.0555 ± 0.0044
		✓	0.2018 ± 0.0106‡	0.1853 ± 0.0156	0.1475 ± 0.0125	0.0561 ± 0.0074
	✓		0.2096 ± 0.0131	0.1792 ± 0.0080‡	0.1458 ± 0.0102	0.0567 ± 0.0093
			0.2048 ± 0.0134	0.1824 ± 0.0031	0.1553 ± 0.0062	0.0589 ± 0.0052
Libras	✓	✓	0.3156 ± 0.0374	0.3162 ± 0.0405	0.4101 ± 0.0302	0.1502 ± 0.0253
		✓	0.2839 ± 0.0032‡	0.2979 ± 0.0079‡	0.3439 ± 0.0107‡	0.1012 ± 0.0057‡
	✓		0.3272 ± 0.0278	0.3460 ± 0.0497	0.4212 ± 0.0300	0.1600 ± 0.0271
			0.2883 ± 0.0069‡	0.2959 ± 0.0148‡	0.3481 ± 0.0125‡	0.1084 ± 0.0080‡
MotorImagery	✓	✓	0.5323 ± 0.0147	0.5271 ± 0.0154	0.0039 ± 0.0034	0.0025 ± 0.0046
		✓	0.5360 ± 0.0178	0.5196 ± 0.0260‡	0.0053 ± 0.0041	0.0041 ± 0.0050
	✓		0.5307 ± 0.0224‡	0.5249 ± 0.0284‡	0.0042 ± 0.0048	0.0032 ± 0.0066
			0.5238 ± 0.0115‡	0.4965 ± 0.0034‡	0.0035 ± 0.0038‡	0.0007 ± 0.0027‡
NATOPS	✓	✓	0.4756 ± 0.0180	0.4638 ± 0.0129	0.4697 ± 0.0383	0.3103 ± 0.0385
		✓	0.4533 ± 0.0457‡	0.4550 ± 0.0494‡	0.4481 ± 0.0457‡	0.2797 ± 0.0460‡
	✓		0.4528 ± 0.0511‡	0.4554 ± 0.0583‡	0.4444 ± 0.0859‡	0.2844 ± 0.0753‡
			0.4750 ± 0.0072‡	0.4797 ± 0.0059	0.4753 ± 0.0118	0.3082 ± 0.0133‡
PEMS-SF	✓	✓	0.3905 ± 0.0724	0.4364 ± 0.0969	0.3625 ± 0.1185	0.2028 ± 0.0839
		✓	0.2768 ± 0.0192‡	0.2752 ± 0.0175‡	0.2469 ± 0.0372‡	0.1170 ± 0.0271‡
	✓		0.4159 ± 0.0708	0.4722 ± 0.0819	0.4329 ± 0.0928	0.2465 ± 0.0908
			0.2550 ± 0.0164‡	0.2553 ± 0.0131‡	0.2111 ± 0.0293‡	0.0900 ± 0.0244‡
PenDigits	✓	✓	0.6796 ± 0.0497	0.6724 ± 0.0488	0.6440 ± 0.0381	0.5249 ± 0.0488
		✓	0.5957 ± 0.0208‡	0.5949 ± 0.0207‡	0.5736 ± 0.0164‡	0.4065 ± 0.0264‡
	✓		0.6614 ± 0.0378‡	0.6457 ± 0.0436‡	0.6188 ± 0.0319‡	0.4953 ± 0.0431‡
			0.5954 ± 0.0262‡	0.5955 ± 0.0250‡	0.5702 ± 0.0197‡	0.4034 ± 0.0284‡
PhonemeSpectra	✓	✓	0.1125 ± 0.0040	0.1119 ± 0.0039	0.1765 ± 0.0084	0.0333 ± 0.0020
		✓	OOM	OOM	OOM	OOM
	✓		0.1036 ± 0.0047‡	0.1044 ± 0.0044‡	0.1576 ± 0.0107‡	0.0277 ± 0.0029‡
			0.1036 ± 0.0016‡	0.1027 ± 0.0021‡	0.1568 ± 0.0039‡	0.0289 ± 0.0014‡
RacketSports	✓	✓	0.5908 ± 0.0704	0.5918 ± 0.0700	0.3431 ± 0.0775	0.2754 ± 0.0886
		✓	0.6330 ± 0.0454	0.6263 ± 0.0497	0.4037 ± 0.0637	0.3443 ± 0.0751
	✓		0.5340 ± 0.0979‡	0.5410 ± 0.0966‡	0.3100 ± 0.1019‡	0.2449 ± 0.1091‡
			0.6026 ± 0.0949	0.6012 ± 0.0964	0.3912 ± 0.1121	0.3290 ± 0.1141
SelfRegulation SCP1	✓	✓	0.7747 ± 0.0140	0.7738 ± 0.0136	0.2371 ± 0.0269	0.3014 ± 0.0296
		✓	0.7647 ± 0.0254‡	0.7635 ± 0.0258‡	0.2224 ± 0.0410‡	0.2816 ± 0.0510‡
	✓		0.7740 ± 0.0170‡	0.7730 ± 0.0160‡	0.2387 ± 0.0412	0.3002 ± 0.0376‡
			0.7647 ± 0.0103‡	0.7644 ± 0.0099‡	0.2150 ± 0.0213‡	0.2794 ± 0.0224‡
SelfRegulation SCP2	✓	✓	0.5253 ± 0.0151	0.5162 ± 0.0079	0.0036 ± 0.0050	0.0010 ± 0.0042
		✓	0.5279 ± 0.0107	0.5270 ± 0.0093	0.0026 ± 0.0022‡	0.0010 ± 0.0029
	✓		0.5147 ± 0.0106‡	0.5141 ± 0.0105‡	0.0010 ± 0.0013‡	0.0013 ± 0.0017
			0.5142 ± 0.0032‡	0.5141 ± 0.0033‡	0.0006 ± 0.0003‡	0.0018 ± 0.0004
Spoken ArabicDigits	✓	✓	0.4260 ± 0.0202	0.4328 ± 0.0168	0.3868 ± 0.0165	0.2365 ± 0.0194
		✓	0.4712 ± 0.0196	0.4778 ± 0.0219	0.4407 ± 0.0153	0.2844 ± 0.0124
	✓		0.4477 ± 0.0191	0.4502 ± 0.0231	0.4457 ± 0.0383	0.2802 ± 0.0395
			0.4637 ± 0.0203	0.4691 ± 0.0204	0.4426 ± 0.0119	0.2821 ± 0.0086
StandWalkJump	✓	✓	0.4000 ± 0.0637	0.3740 ± 0.0724	0.0441 ± 0.0480	0.0171 ± 0.0522
		✓	0.4000 ± 0.0277	0.3749 ± 0.0372	0.0458 ± 0.0276	0.0173 ± 0.0277
	✓		0.4222 ± 0.0502	0.3944 ± 0.0506	0.0417 ± 0.0306‡	0.0215 ± 0.0349
			0.4148 ± 0.0544	0.3852 ± 0.0626	0.0547 ± 0.0419	0.0341 ± 0.0161
UWaveGesture Library	✓	✓	0.4164 ± 0.0435	0.4043 ± 0.0445	0.3456 ± 0.0479	0.1996 ± 0.0452
		✓	0.3373 ± 0.0086‡	0.3156 ± 0.0090‡	0.1918 ± 0.0200‡	0.0954 ± 0.0114‡
	✓		0.3945 ± 0.0411‡	0.3881 ± 0.0581‡	0.2768 ± 0.0378‡	0.1487 ± 0.0261‡
			0.3127 ± 0.0232‡	0.2967 ± 0.0177‡	0.1782 ± 0.0133‡	0.0860 ± 0.0156‡

clustering. Temporal restoration and prototype-guided refinement are both required after source-level pruning.

The complete ablations therefore support the role assignment summarized in the main paper. CCMA restricts which patches participate in early feature interaction, MEVG organizes the admitted content into complementary endogenous views, reconstruction keeps the pruned representation semantically traceable, and contrastive learning links the organized views to cluster geometry. SIGMA therefore does not rely on a single dominant block. Its empirical gains come from a compact chain in which admission, view organization, semantic restoration, and cluster refinement address different failure modes of redundancy-dominated MTS clustering. This role assignment helps disentangle redundancy suppression from semantic preservation, then connects retained semantics to cluster refinement under the prune-and-preserve framework.

C. Additional Training-Loss Curves

The main paper evaluates training stability on four representative datasets. This supplemental material subsection extends the same diagnosis to DuckDuckGeese, EthanolConcentration, FaceDetection, Heartbeat, JapaneseVowels, and PEMS-SF, covering both weak label-to-signal correspondence and clearer cluster-revealing structures. Fig. 2 reports the mean loss over five seeds, with shaded bands denoting standard deviation. Across the six datasets, the warm-up phase first reduces the reconstruction-dominated objective, indicating that admitted patches obtain a stable temporal-content anchor after source-level admission. This stage is important because pruning shortens the encoder input and may remove contextual relations needed by later clustering objectives.

After the dual-route ramp-up begins, several curves show a short increase or fluctuation as the objective expands from semantic anchoring to inter-view complementarity and prototype-level alignment. The relevant signal is therefore bounded optimization after $\mathcal{L}_{\rm orth}$ and $\mathcal{L}_{\rm pa}$ enter training, instead of a strictly monotonic transition. In later epochs, DuckDuckGeese, FaceDetection, Heartbeat, JapaneseVowels, and PEMS-SF move into smoother refinement, and Ethanol-

TABLE V
ABLATION STUDIES ON ROUTE DESIGNS OF SIGMA, PART I. RED "‡" MARKS DEGRADATION FROM COMPLETE SIGMA.

Datasets (1-15)	Components		Metrics			
	Recon	Contra	ACC	F1	NMI	ARI
ArticularyWord Recognition	✓	✓	0.7329 ± 0.0395	0.7696 ± 0.0443	0.8037 ± 0.0246	0.6029 ± 0.0470
		✓	0.6035 ± 0.0317‡	0.6212 ± 0.0261‡	0.7016 ± 0.0278‡	0.4614 ± 0.0399‡
	✓		0.8247 ± 0.0562	0.8514 ± 0.0558	0.8747 ± 0.0426	0.7445 ± 0.0793
			0.1419 ± 0.0485‡	0.2366 ± 0.0323‡	0.2485 ± 0.0639‡	0.0353 ± 0.0299‡
AtrialFibrillation	✓	✓	0.4933 ± 0.0327	0.4720 ± 0.0232	0.1215 ± 0.0474	0.0443 ± 0.0492
		✓	0.4933 ± 0.0249	0.4763 ± 0.0291	0.0833 ± 0.0242‡	0.0169 ± 0.0203‡
	✓		0.4733 ± 0.0389‡	0.4520 ± 0.0542‡	0.0765 ± 0.0312‡	0.0190 ± 0.0333‡
			0.4600 ± 0.0389‡	0.4518 ± 0.0598‡	0.0937 ± 0.0356‡	0.0212 ± 0.0325‡
BasicMotions	✓	✓	0.9675 ± 0.0150	0.9671 ± 0.0154	0.9273 ± 0.0277	0.9169 ± 0.0363
		✓	0.9050 ± 0.1226‡	0.9319 ± 0.0784‡	0.9161 ± 0.0911‡	0.8633 ± 0.1575‡
	✓		0.6600 ± 0.0366‡	0.8105 ± 0.0355‡	0.7754 ± 0.0499‡	0.6063 ± 0.0482‡
			0.5100 ± 0.0289‡	0.6355 ± 0.0519‡	0.5518 ± 0.0478‡	0.3927 ± 0.0402‡
Character Trajectories	✓	✓	0.7879 ± 0.0499	0.8091 ± 0.0408	0.8463 ± 0.0284	0.7214 ± 0.0576
		✓	0.6714 ± 0.0832‡	0.7199 ± 0.0942‡	0.7443 ± 0.0645‡	0.5830 ± 0.0923‡
	✓		0.7389 ± 0.0708‡	0.7674 ± 0.0674‡	0.8189 ± 0.0503‡	0.6750 ± 0.0821‡
			0.0651 ± 0.0000‡	0.1222 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
Cricket	✓	✓	0.8244 ± 0.0255	0.8463 ± 0.0115	0.8522 ± 0.0167	0.7259 ± 0.0299
		✓	0.7900 ± 0.1095‡	0.8001 ± 0.1052‡	0.8279 ± 0.0699‡	0.6912 ± 0.1158‡
	✓		0.7511 ± 0.0708‡	0.7828 ± 0.0339‡	0.7682 ± 0.0476‡	0.5964 ± 0.0836‡
			0.1000 ± 0.0333‡	0.2018 ± 0.0960‡	0.0721 ± 0.1442‡	0.0121 ± 0.0241‡
DuckDuckGeese	✓	✓	0.3900 ± 0.0141	0.3730 ± 0.0193	0.1749 ± 0.0375	0.0795 ± 0.0345
		✓	0.4740 ± 0.0546	0.4690 ± 0.0553	0.2439 ± 0.0457	0.1618 ± 0.0523
	✓		0.3880 ± 0.0306‡	0.3682 ± 0.0377‡	0.1562 ± 0.0375‡	0.0621 ± 0.0248‡
			0.2000 ± 0.0000‡	0.3333 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
ERing	✓	✓	0.7307 ± 0.0629	0.7287 ± 0.0680	0.5996 ± 0.0590	0.5124 ± 0.0760
		✓	0.7533 ± 0.0419	0.7553 ± 0.0421	0.6735 ± 0.0240	0.5854 ± 0.0353
	✓		0.8020 ± 0.0937	0.8006 ± 0.0951	0.7408 ± 0.0702	0.6688 ± 0.1070
			0.1667 ± 0.0000‡	0.2857 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
EigenWorms	✓	✓	0.4193 ± 0.0788	0.3646 ± 0.0438	0.1612 ± 0.0594	0.1515 ± 0.0726
		✓	0.4556 ± 0.0523	0.4150 ± 0.0632	0.2774 ± 0.0369	0.2479 ± 0.0381
	✓		0.3382 ± 0.0317‡	0.3009 ± 0.0439‡	0.0843 ± 0.0295‡	0.0512 ± 0.0158‡
			0.4247 ± 0.0000	0.5962 ± 0.0000	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
Epilepsy	✓	✓	0.4655 ± 0.0229	0.4573 ± 0.0233	0.2355 ± 0.0362	0.1668 ± 0.0336
		✓	0.5716 ± 0.0429	0.5746 ± 0.0356	0.3998 ± 0.0509	0.3182 ± 0.0499
	✓		0.4320 ± 0.0292‡	0.4259 ± 0.0276‡	0.2047 ± 0.0333‡	0.1442 ± 0.0279‡
			0.4291 ± 0.0332‡	0.4217 ± 0.0355‡	0.1670 ± 0.0517‡	0.1224 ± 0.0370‡
Ethanol Concentration	✓	✓	0.3061 ± 0.0208	0.2766 ± 0.0066	0.0199 ± 0.0156	0.0108 ± 0.0123
		✓	0.2916 ± 0.0080‡	0.2841 ± 0.0078	0.0129 ± 0.0041‡	0.0032 ± 0.0014‡
	✓		0.3115 ± 0.0143	0.2799 ± 0.0252	0.0308 ± 0.0187	0.0144 ± 0.0106
			0.3179 ± 0.0130	0.3277 ± 0.0507	0.0442 ± 0.0192	0.0238 ± 0.0127
FaceDetection	✓	✓	0.5107 ± 0.0042	0.5100 ± 0.0049	0.0004 ± 0.0003	0.0004 ± 0.0004
		✓	0.5090 ± 0.0055‡	0.5086 ± 0.0056‡	0.0003 ± 0.0003‡	0.0003 ± 0.0004‡
	✓		0.5000 ± 0.0000‡	0.6667 ± 0.0000	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
			0.5000 ± 0.0000‡	0.6667 ± 0.0000	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
FingerMovements	✓	✓	0.5188 ± 0.0085	0.5142 ± 0.0086	0.0013 ± 0.0007	0.0006 ± 0.0010
		✓	0.5135 ± 0.0097‡	0.5114 ± 0.0111‡	0.0008 ± 0.0009‡	0.0013 ± 0.0012
	✓		0.5231 ± 0.0102	0.5188 ± 0.0102	0.0019 ± 0.0012	0.0002 ± 0.0016‡
			0.5000 ± 0.0000‡	0.6667 ± 0.0000	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
HandMovement Direction	✓	✓	0.3333 ± 0.0143	0.3298 ± 0.0171	0.0205 ± 0.0063	0.0060 ± 0.0052
		✓	0.3077 ± 0.0162‡	0.3061 ± 0.0171‡	0.0148 ± 0.0062‡	0.0005 ± 0.0053‡
	✓		0.3111 ± 0.0208‡	0.3077 ± 0.0221‡	0.0215 ± 0.0075	0.0053 ± 0.0072‡
			0.3068 ± 0.0068‡	0.2901 ± 0.0134‡	0.0201 ± 0.0043‡	0.0021 ± 0.0020‡
Handwriting	✓	✓	0.2386 ± 0.0181	0.2433 ± 0.0195	0.3461 ± 0.0286	0.1006 ± 0.0180
		✓	0.1706 ± 0.0061‡	0.1741 ± 0.0083‡	0.2227 ± 0.0045‡	0.0385 ± 0.0019‡
	✓		0.2054 ± 0.0071‡	0.2071 ± 0.0059‡	0.2809 ± 0.0091‡	0.0663 ± 0.0054‡
			0.1428 ± 0.0060‡	0.1408 ± 0.0057‡	0.1813 ± 0.0062‡	0.0217 ± 0.0024‡
Heartbeat	✓	✓	0.6636 ± 0.0589	0.5679 ± 0.0507	0.0391 ± 0.0232	0.0689 ± 0.0404
		✓	0.5628 ± 0.0376‡	0.5053 ± 0.0802‡	0.0459 ± 0.0155	0.0065 ± 0.0412‡
	✓		0.7002 ± 0.0176	0.7537 ± 0.1522	0.0170 ± 0.0068‡	0.0161 ± 0.0034‡
			0.7213 ± 0.0000	0.8381 ± 0.0000	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡

TABLE VI
ABLATION STUDIES ON ROUTE DESIGNS OF SIGMA, PART II. RED "‡" MARKS DEGRADATION FROM COMPLETE SIGMA.

Datasets (16-30)	Components		Metrics			
	Recon	Contra	ACC	F1	NMI	ARI
InsectWingbeat	✓	✓	0.1344 ± 0.0023	0.1202 ± 0.0033	0.0153 ± 0.0012	0.0061 ± 0.0007
		✓	0.1206 ± 0.0063‡	0.1104 ± 0.0111‡	0.0053 ± 0.0027‡	0.0025 ± 0.0014‡
	✓		0.1000 ± 0.0000‡	0.1818 ± 0.0000	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
			0.1000 ± 0.0000‡	0.1818 ± 0.0000	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
JapaneseVowels	✓	✓	0.8647 ± 0.0447	0.8626 ± 0.0432	0.8195 ± 0.0260	0.7682 ± 0.0565
		✓	0.8859 ± 0.0310	0.8942 ± 0.0270	0.8224 ± 0.0315	0.7627 ± 0.0522‡
	✓		0.6441 ± 0.1450‡	0.6632 ± 0.1467‡	0.5833 ± 0.1473‡	0.4328 ± 0.1564‡
			0.1844 ± 0.0000‡	0.3113 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
LSST	✓	✓	0.2042 ± 0.0087	0.1805 ± 0.0063	0.1456 ± 0.0063	0.0555 ± 0.0044
		✓	0.3362 ± 0.0017	0.2228 ± 0.0094	0.1699 ± 0.0142	0.0714 ± 0.0062
	✓		0.1969 ± 0.0139‡	0.1776 ± 0.0102‡	0.1427 ± 0.0084‡	0.0537 ± 0.0075‡
			0.3155 ± 0.0000	0.4797 ± 0.0000	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
Libras	✓	✓	0.3156 ± 0.0374	0.3162 ± 0.0405	0.4101 ± 0.0302	0.1502 ± 0.0253
		✓	0.3600 ± 0.0283	0.3572 ± 0.0231	0.4518 ± 0.0285	0.1916 ± 0.0259
	✓		0.4161 ± 0.0210	0.4303 ± 0.0315	0.5402 ± 0.0261	0.2622 ± 0.0231
			0.2644 ± 0.0235‡	0.2673 ± 0.0222‡	0.3222 ± 0.0456‡	0.0927 ± 0.0307‡
MotorImagery	✓	✓	0.5323 ± 0.0147	0.5271 ± 0.0154	0.0039 ± 0.0034	0.0025 ± 0.0046
		✓	0.5079 ± 0.0044‡	0.4726 ± 0.0036‡	0.0004 ± 0.0004‡	0.0016 ± 0.0004‡
	✓		0.5270 ± 0.0074‡	0.5048 ± 0.0179‡	0.0034 ± 0.0023‡	0.0009 ± 0.0017‡
			OOM	OOM	OOM	OOM
NATOPS	✓	✓	0.4756 ± 0.0180	0.4638 ± 0.0129	0.4697 ± 0.0383	0.3103 ± 0.0385
		✓	0.4972 ± 0.0478	0.4940 ± 0.0434	0.4796 ± 0.0594	0.3245 ± 0.0508
	✓		0.4328 ± 0.0428‡	0.4242 ± 0.0431‡	0.3578 ± 0.0955‡	0.2377 ± 0.0650‡
			0.1967 ± 0.0600‡	0.3526 ± 0.1337‡	0.0770 ± 0.1539‡	0.0221 ± 0.0441‡
PEMS-SF	✓	✓	0.3905 ± 0.0724	0.4364 ± 0.0969	0.3625 ± 0.1185	0.2028 ± 0.0839
		✓	0.2368 ± 0.0216‡	0.2476 ± 0.0355‡	0.1705 ± 0.0511‡	0.0609 ± 0.0218‡
	✓		0.2700 ± 0.0273‡	0.2785 ± 0.0369‡	0.1930 ± 0.0781‡	0.0771 ± 0.0412‡
			0.1477 ± 0.0000‡	0.2574 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
PenDigits	✓	✓	0.6796 ± 0.0497	0.6724 ± 0.0488	0.6440 ± 0.0381	0.5249 ± 0.0488
		✓	0.7139 ± 0.0446	0.7030 ± 0.0448	0.6472 ± 0.0277	0.5420 ± 0.0420
	✓		0.1041 ± 0.0000‡	0.1885 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
			0.1041 ± 0.0000‡	0.1885 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
PhonemeSpectra	✓	✓	0.1125 ± 0.0040	0.1119 ± 0.0039	0.1765 ± 0.0084	0.0333 ± 0.0020
		✓	0.0985 ± 0.0017‡	0.0967 ± 0.0012‡	0.1443 ± 0.0051‡	0.0243 ± 0.0012‡
	✓		0.0256 ± 0.0000‡	0.0500 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
			0.0256 ± 0.0000‡	0.0500 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
RacketSports	✓	✓	0.5908 ± 0.0704	0.5918 ± 0.0700	0.3431 ± 0.0775	0.2754 ± 0.0886
		✓	0.6389 ± 0.0424	0.6436 ± 0.0345	0.4214 ± 0.0229	0.3526 ± 0.0483
	✓		0.4488 ± 0.0172‡	0.4449 ± 0.0185‡	0.1929 ± 0.0490‡	0.1187 ± 0.0352‡
			0.4244 ± 0.0367‡	0.3947 ± 0.0725‡	0.1550 ± 0.0234‡	0.1061 ± 0.0245‡
Self RegulationSCP1	✓	✓	0.7747 ± 0.0140	0.7738 ± 0.0136	0.2371 ± 0.0269	0.3014 ± 0.0296
		✓	0.7458 ± 0.0132‡	0.7421 ± 0.0156‡	0.2011 ± 0.0274‡	0.2411 ± 0.0261‡
	✓		0.7337 ± 0.0714‡	0.7246 ± 0.0859‡	0.1956 ± 0.0777‡	0.2376 ± 0.1041‡
			0.7480 ± 0.0175‡	0.7477 ± 0.0176‡	0.1878 ± 0.0261‡	0.2458 ± 0.0341‡
Self RegulationSCP2	✓	✓	0.5253 ± 0.0151	0.5162 ± 0.0079	0.0036 ± 0.0050	0.0010 ± 0.0042
		✓	0.5347 ± 0.0139	0.5325 ± 0.0152	0.0041 ± 0.0021	0.0030 ± 0.0029
	✓		0.5158 ± 0.0173‡	0.5106 ± 0.0157‡	0.0017 ± 0.0031‡	0.0003 ± 0.0040‡
			0.5200 ± 0.0061‡	0.5191 ± 0.0059	0.0013 ± 0.0007‡	0.0009 ± 0.0009‡
Spoken ArabicDigits	✓	✓	0.4260 ± 0.0202	0.4328 ± 0.0168	0.3868 ± 0.0165	0.2365 ± 0.0194
		✓	0.3631 ± 0.0512‡	0.3642 ± 0.0468‡	0.3212 ± 0.0486‡	0.1523 ± 0.0453‡
	✓		0.1000 ± 0.0000‡	0.1819 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
			0.1000 ± 0.0000‡	0.1819 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
StandWalkJump	✓	✓	0.4000 ± 0.0637	0.3740 ± 0.0724	0.0441 ± 0.0480	0.0171 ± 0.0522
		✓	0.4593 ± 0.0378	0.4537 ± 0.0361	0.0530 ± 0.0194	0.0194 ± 0.0251
	✓		0.4519 ± 0.0363	0.4355 ± 0.0497	0.0543 ± 0.0236	0.0178 ± 0.0199
			0.4296 ± 0.0181	0.3740 ± 0.0214‡	0.0924 ± 0.0506	0.0016 ± 0.0243‡
UWaveGesture Library	✓	✓	0.4164 ± 0.0435	0.4043 ± 0.0445	0.3456 ± 0.0479	0.1996 ± 0.0452
		✓	0.4023 ± 0.0342‡	0.3992 ± 0.0359‡	0.2887 ± 0.0371‡	0.1684 ± 0.0280‡
	✓		0.6318 ± 0.0456	0.6520 ± 0.0381	0.5779 ± 0.0160	0.4497 ± 0.0166
			0.1250 ± 0.0000‡	0.2222 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡

Concentration also stabilizes after a brief high-variance interval. The bounded shaded bands indicate that prototype-guided feedback does not cause persistent oscillation or collapse. These curves support the staged design of SIGMA, where reconstruction first anchors retained temporal content, inter-view orthogonality preserves complementary endogenous views, and prototype alignment refines the fused cluster space under the prune-and-preserve pipeline.

D. Scalability under Increasing Mask Rates

The main paper answers Q3 through runtime and memory comparisons under increasing sample size, sequence length, and input dimension. This supplemental material subsection further verifies whether the efficiency gain follows the intended admission mechanism. When cost reduction comes from pruning before encoder-side feature interaction, a higher mask rate should yield lower normalized execution time. Fig. 3 examines this relation under varying N, T, and D. Given patch length ℓ_{p}, each sequence is divided into $m = \lceil t/\ell_{\mathrm{p}} \rceil$ patches. Under retained ratio r_{a}, CCMA keeps $m_{\mathrm{a}} = \lceil r_{\mathrm{a}} m \rceil$ patches for encoding, and the mask rate is obtained as:

$$r_{\mathrm{m}} = 1 - \frac{m_{\mathrm{a}}}{m}. \tag{1}$$

The experiment increases r_{m} from 0% to 90% by decreasing r_{a}, and normalizes execution time by the zero-mask cost to compare relative savings across scale factors.

Fig. 3 shows a consistent runtime decrease as r_{m} increases. The average reduction reaches 78.2% under varying N, 86.3% under varying T, and 78.8% under varying D. The strongest gain appears with longer sequences, matching the complexity analysis because the dominant attention term depends on the retained patch number m_{a}. As r_{m} grows, fewer admitted patches produce fewer pairwise interactions inside the encoder. The same trend also holds when more samples accumulate the saving or when higher-dimensional projections operate on fewer tokens. These results confirm that CCMA performs source-side, computation-aware pruning by changing the ad-

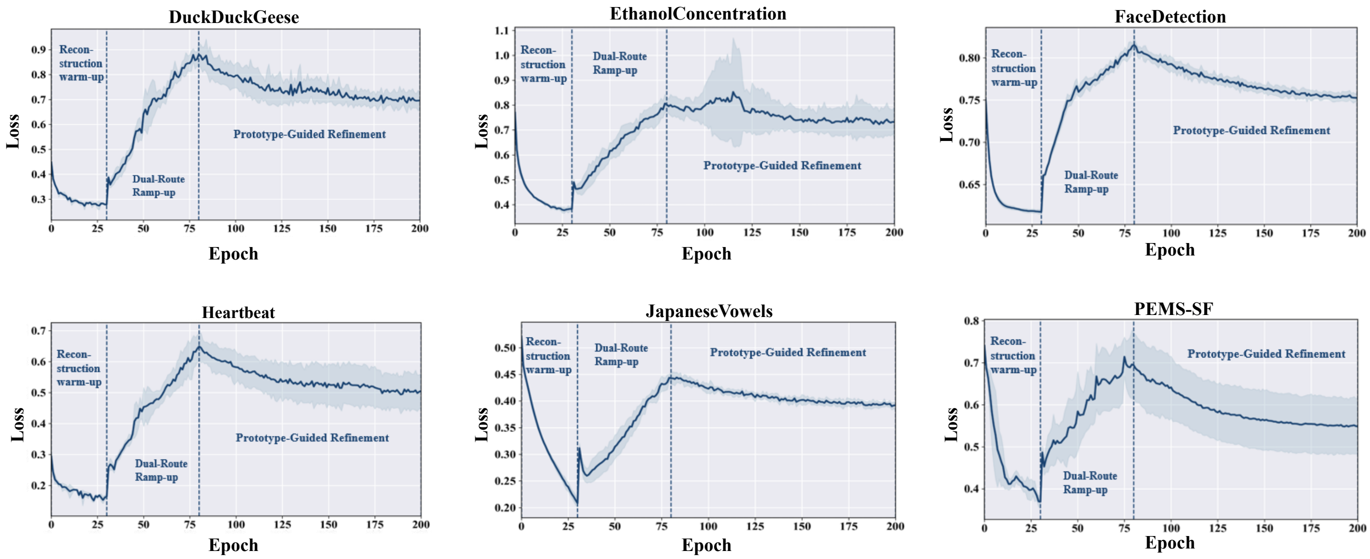

Fig. 2. Additional training-loss curves during SIGMA training on six UEA datasets. Each curve reports the mean loss across five random seeds, and the shaded band denotes standard deviation. Vertical dashed lines mark the semantic warm-up, dual-route ramp-up, and prototype-guided refinement stages.

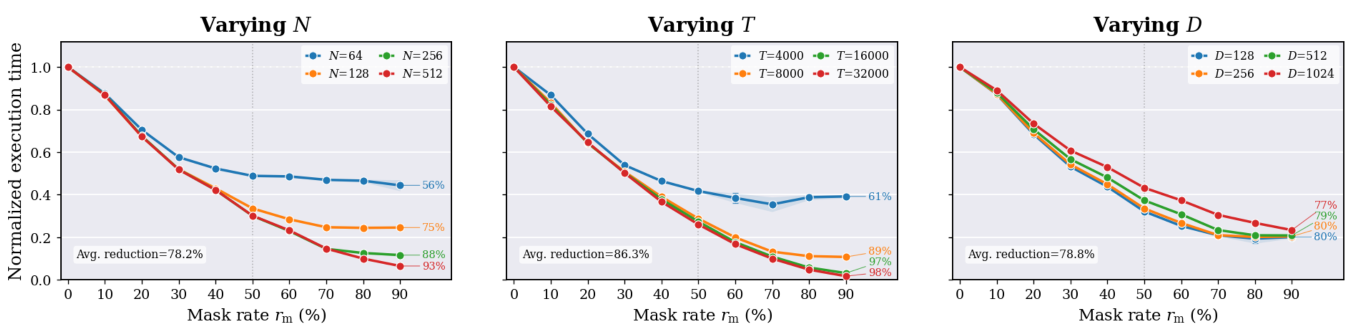

Fig. 3. Scalability under increasing mask rates. The three panels vary N, T, and D, respectively. For each setting, execution time is normalized by the cost at zero mask rate. Shaded bands denote standard deviation, and endpoint annotations report the reduction at 90% mask rate.

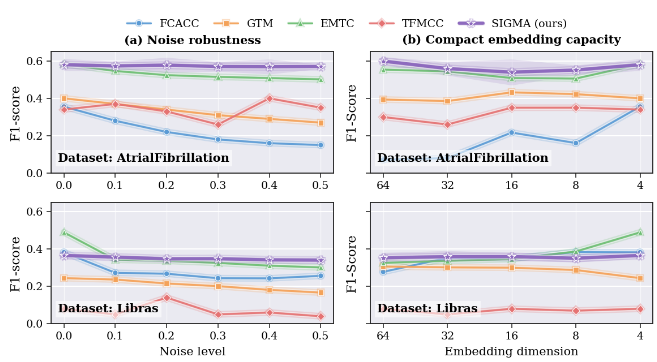

Fig. 4. **Robustness and compactness diagnostics.** F1-score on AtrialFibrillation and Libras under (a) increasing Gaussian noise and (b) reduced embedding dimensions from 64 to 4. SIGMA keeps stable accuracy in both settings, showing that retained representations remain usable under perturbation and compact latent budgets.

missible input set before redundant patches shape feature interaction.

E. Robustness and Compactness Diagnostics

Fig. 4 supplements Q4 by testing whether the retained representation remains usable under input perturbation and compact latent budgets. In Fig. 4(a), Gaussian noise is added to AtrialFibrillation and Libras. SIGMA keeps a smooth F1 trajectory as the noise level increases, especially on AtrialFibrillation, where the curve stays above the compared baselines. On Libras, several methods fluctuate or decline after perturbation, whereas SIGMA remains stable across noise levels. The result suggests that confidence-calibrated admission filters part of the corrupted or routine content before feature interaction, and the retained views still provide sufficient context for clustering.

Fig. 4(b) further reduces the embedding dimension from 64 to 4. SIGMA maintains competitive F1-score under compressed dimensions on both datasets, showing that the learned cluster structure does not rely on an oversized latent space. EMTC remains competitive on Libras at the smallest dimension, but SIGMA gives steadier behavior across the full dimension range. Together, the two diagnostics indicate that the prune-but-preserve design supports perturbation tolerance and compact representation learning, which is essential for efficient MTS organization under constrained computation and storage budgets.

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